

Government College of Engineering Aurangabad, Chhatrapati Sambhaji Nagar

(An Autonomous Institute of Government of Maharashtra)

End Semester Examination May-June 2024

First Year MCA – NEP – Semester – II

MCPCC1002 Soft Computing

Time: 2.30 Hours

17 MAY 2024

Max. Marks: 60

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. Attempt all questions.
2. Figures to the right indicate full marks.
3. Assume suitable data if necessary and state it clearly.
4. Use of Non-Programmable Calculator and Data Sheet is allowed.

Q.1	Attempt any two	CO	B.T. Level	Marks
A	What is soft Computing? Compare hard and soft computing. List various soft computing techniques.	1	K2	06
B	Draw biological NN and explain its parts? Draw neat sketch of biological equivalent NN and explain various parts.	1	K2	06
R2 C	State and explain few applications of ANNs which are used to solve some real life problems.	1	K2	06
Q.2	Attempt any two	CO	B.T. Level	Marks
R4 A	What are linear and non-linearly separable problems? How it is classified using various algorithms using ANN.	2	K2	06
R3 B	State and explain perceptron learning algorithm	2	K3	06
C	Obtain the output of the neuron using binary and bipolar sigmoidal function: $[x_1, x_2, x_3] = [0.8, 0.6, 0.4]$, $[w_1, w_2, w_3] = [0.1, 0.3, -0.2]$, bias is $b = 0.35$	2	K3	06
Q.3	Attempt any two	CO	B.T. Level	Marks
A	Write and explain Perceptron Training Algorithm for Multiple Output Classes.	3	K2	06
B	Find the weights required to perform the following classification using perceptron network. The vectors $(1, 1, 1, 1)$ and $(-1, 1, -1, -1)$ are belonging to the class (so have target value 1), vectors $(1, 1, 1, -1)$ and $(1, -1, -1, 1)$ are not belonging to the class (so have target value -1). Assume learning rate as 1 and initial weights as 0.	3	K3	06
C	Using Madaline network, implement XOR function with bipolar inputs and targets. Assume the required parameters for training of the network.	3	K3	06

		CO	B.T. Level	Marks												
Q.4	Attempt any two															
A	Explain the algorithm for learning vector quantization with suitable data.	4	K3	06												
B	Consider a Kohonen self-organizing net with two cluster units and five input units. The weight vectors for the cluster units are given by $W_1 = [1.0 \ 0.9 \ 0.7 \ 0.5 \ 0.3]$ $w_2 = [0.3 \ 0.5 \ 0.7 \ 0.9 \ 1.0]$ Use the square of the Euclidean distance to find the winning cluster unit for the input pattern $x = [0.0 \ 0.5 \ 1.0 \ 0.5 \ 0.0]$. Using a learning rate of 0.25, find the new weights for the winning unit.	4	K3	06												
C	Construct and test an LVQ net with five vectors assigned to two classes. The given vectors along with the classes are as shown <table border="1"> <thead> <tr> <th>Vector</th> <th>Class</th> </tr> </thead> <tbody> <tr> <td>[0 0 1 1]</td> <td>1</td> </tr> <tr> <td>[1 0 0 0]</td> <td>2</td> </tr> <tr> <td>[0 0 0 1]</td> <td>2</td> </tr> <tr> <td>[1 1 0 0]</td> <td>1</td> </tr> <tr> <td>[0 1 1 0]</td> <td>1</td> </tr> </tbody> </table>	Vector	Class	[0 0 1 1]	1	[1 0 0 0]	2	[0 0 0 1]	2	[1 1 0 0]	1	[0 1 1 0]	1	4	K3	06
Vector	Class															
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[1 0 0 0]	2															
[0 0 0 1]	2															
[1 1 0 0]	1															
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Q.5	Attempt any two	CO	B.T. Level	Marks												
A	What are fuzzy sets? State and explain the properties and operations on fuzzy sets.	5	K2	06												
B	Using the inference approach, find the membership values for the triangular shapes L, R, E, LR , and I for a triangle with angles $45^\circ, 55^\circ$ and 80° .	5	K3	06												
C	For the two given fuzzy sets $A = \left\{ \frac{0.1}{0} + \frac{0.2}{1} + \frac{0.4}{2} + \frac{0.6}{3} + \frac{1}{4} \right\}$ $B = \left\{ \frac{1}{0} + \frac{0.5}{1} + \frac{0.7}{2} + \frac{0.3}{3} + \frac{0}{4} \right\}$ find the following: (a) $A \cup B$ (b) $A \cap B$ (c) $\bar{A}$ (d) $\bar{B}$ (e) $A - B$ (f) $B - A$	5	K3	06												

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level